	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering** An Autonomous Institute (Permanently affiliated to Savitribai Phule Pune University)	SEMESTER- VII

Assignment 2

Create, configure, connect to, monitor, and terminate a virtual machine instance using the selected cloud platform (AWS EC2 or Azure Virtual Machine)

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1. Aim : Create, configure, connect to, monitor, and terminate a virtual machine instance using the selected cloud platform (AWS EC2 or Azure Virtual Machine)

2. Objectives

- To Study the various computing services in the Amazon AWS cloud.
- To study the concept of Virtual Machine
- To study create and configure EC2 instance

3. Theory

i) Cloud and AWS

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and intelligence over the Internet instead of

using local servers or personal computers.
In traditional computing, organizations had to buy and maintain physical hardware such as servers and networking devices. In cloud computing, these resources are provided by cloud service providers and can be accessed anytime through the internet.

Key Characteristics of Cloud Computing

1. **On-Demand Self-Service** – Users can access resources whenever needed without human interaction with the service provider.

2. **Broad Network Access** – Services are available over the internet and accessible from various devices.

3. **Resource Pooling** – Computing resources are shared among multiple users.

4. **Rapid Elasticity** – Resources can be scaled up or down quickly.

5. **Measured Service** – Users pay only for what they use (pay-as-you-go model).

Amazon Web Services (AWS) provides a wide range of services for computing, storage, networking, database management, machine learning, and more.

Common Applications of AWS

1. **Website Hosting** – Hosting static and dynamic websites.

2. **Application Development** – Deploying web and mobile applications.

3. **Data Storage and Backup** – Using services like S3 for storing data.

4. **Big Data and Analytics** – Processing large datasets.

5. **Machine Learning and AI** – Building intelligent applications.
6. **Virtual Servers** – Creating and managing EC2 instances.

7. **DevOps and CI/CD** – Automating software deployment processes.

ii) Virtual machines

A Virtual Machine (VM) is a software-based simulation of a physical computer. It behaves like a real computer with its own operating system, CPU, memory, and storage. Instead of running directly on hardware, a VM runs on a virtualization layer called a hypervisor.

Types of Hypervisors

1. **Type 1 (Bare Metal)** – Runs directly on hardware (e.g., VMware ESXi, Microsoft Hyper-V).
2. **Type 2 (Hosted)** – Runs on top of an operating system (e.g., VirtualBox, VMware Workstation).

Advantages of Virtual Machines

- Efficient use of hardware
- Isolation between systems
- Easy backup and migration
- Cost-effective
- Scalability

In AWS, EC2 instances are virtual machines running on Amazon's physical servers.

iii) Steps to launch EC2 instance

Amazon EC2 is a web service that provides resizable virtual servers in the cloud.
It allows users to:

- Launch virtual machines (instances)
- Choose operating systems (Linux, Windows, etc.)
- Configure CPU, RAM, and storage
- Control network access
- Start, stop, or terminate instances

Step 1: Login to AWS Console

- Open https://aws.amazon.com
- Login using AWS account credentials.
- Go to AWS Management Console.

Step 2: Navigate to EC2 Dashboard

- In the search bar, type EC2.
- Click on EC2 under Services and open the dashboard.

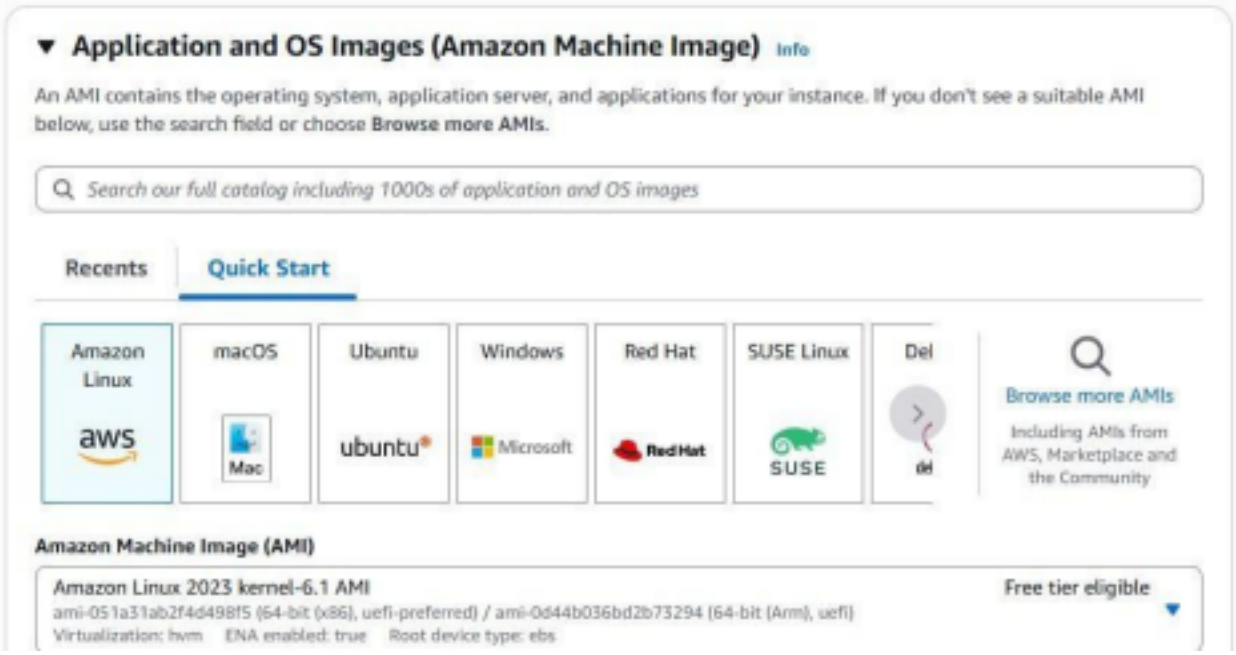
Step 3: Launch Instance

- Click on the Launch Instance button.

Step 4: Choose Amazon Machine Image

(AMI) ● Select an operating system.

An AMI is a template that contains OS and pre-installed software.



Step 5: Choose Instance Type

- Select instance type based on requirements.

Instance type defines: CPU, RAM, Performance



Step 6: Configure Key Pair

- Create a new key pair or use an existing one and download the .pem file.

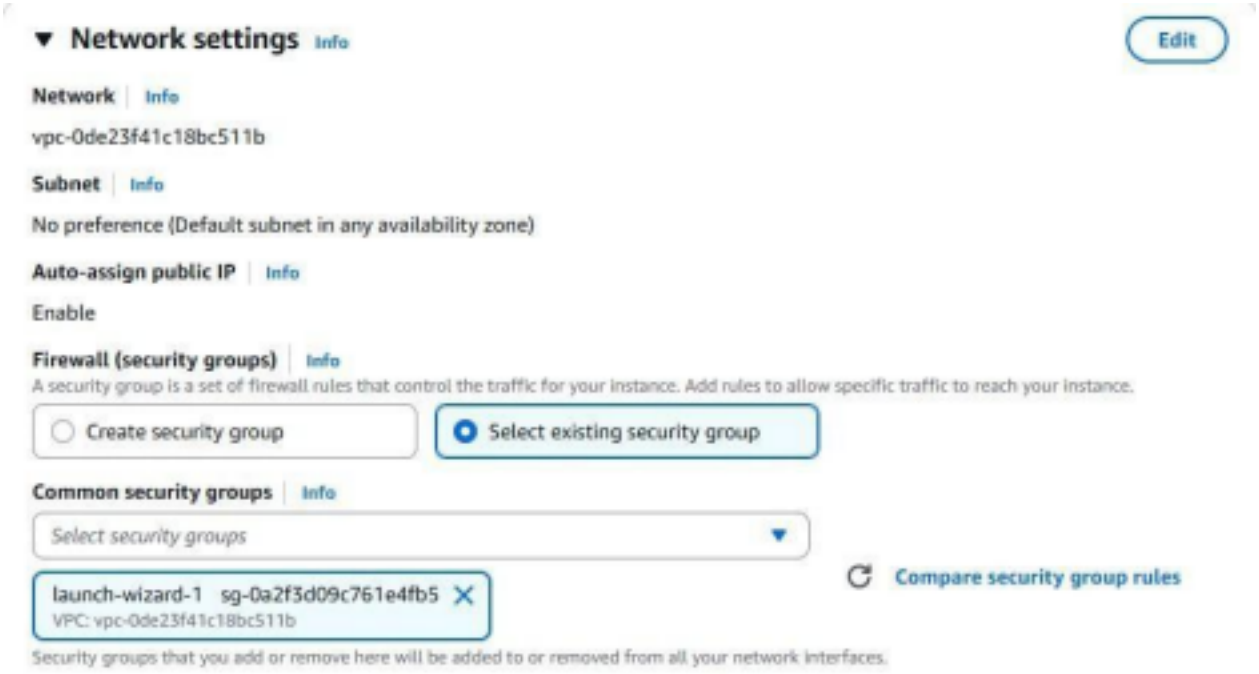
This key is used to securely connect to the instance using SSH.



Step 7: Configure Network Settings

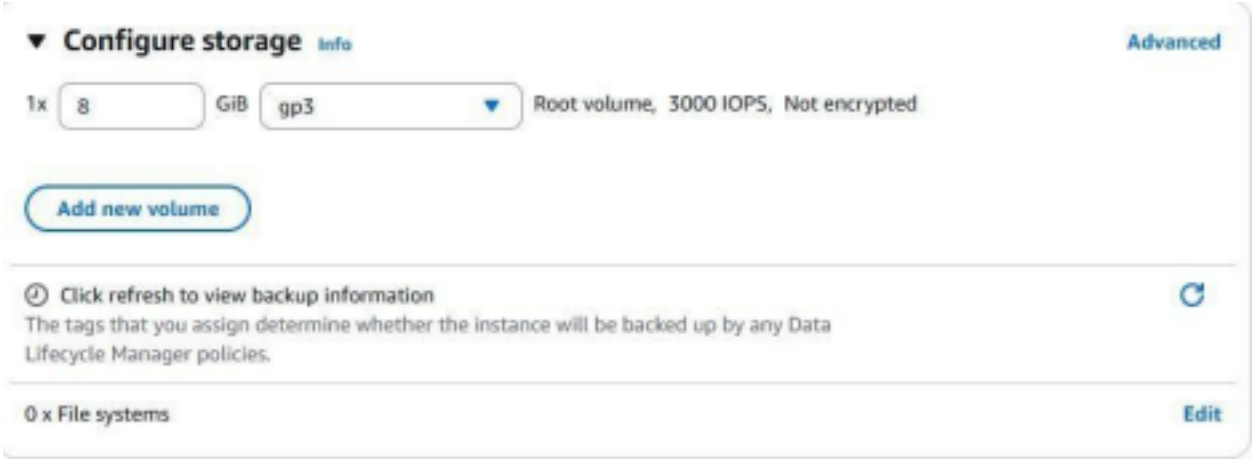
- Select or create a Security Group.
- Add inbound rules:
    - SSH (Port 22) for Linux
    - HTTP (Port 80) for web server
    - HTTPS (Port 443)

Security groups act as virtual firewalls.



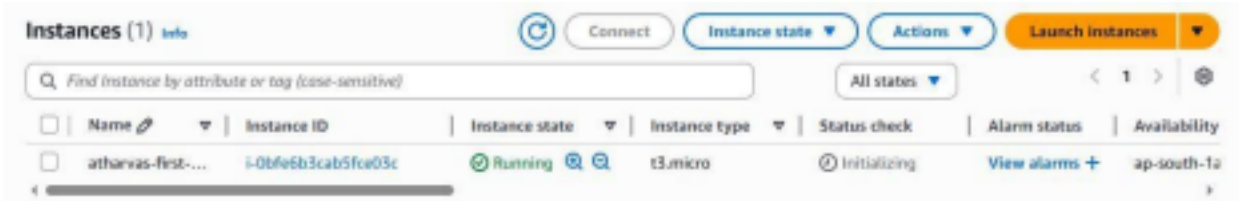
Step 8: Configure Storage

- Add storage if needed.



Step 9: Review and Launch

- Review all configurations and then click Launch Instance.



4. Conclusion

The assignment helped understand the concept of cloud computing and its service models: IaaS, PaaS, and SaaS. Understood the concept of Virtual Machines and how they function using virtualization technology.

Further, explored Amazon Web Services (AWS), and learned about Amazon EC2 as an Infrastructure as a Service (IaaS) offering. Successfully studied the step-by-step procedure to launch and configure an EC2 instance, including selecting an AMI, instance type, security settings, storage, and connecting via SSH.